Ilesh Patel

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Professional Summary

Results-driven IT student specializing in Artificial Intelligence and Machine Learning, with strong skills in Python, Data Structures, and Database Systems. Experienced in Data Preprocessing, Exploratory Data Analysis (EDA), and Feature Engineering, and building ML models using Scikit-learn, Pandas, and NumPy. Knowledgeable in Deep Learning (CNNs, RNNs), Natural Language Processing (NLP), and Generative AI with LLM Integration. Focused on developing End-to-End Scalable AI Solutions that deliver measurable real-world impact.

Skills

Artificial Intelligence & Machine Learning

- Deep Learning
- Neural Networks, Transformers
- Large Language Models (LLMs)
- NLP & Semantic Search
- Embeddings & Vector Similarity

Generative AI Systems

- Context Management & Chunking Strategies
- Multi-step Reasoning Pipelines
- AI Agents & Tool-calling Workflows
- Structured Output Generation

Frameworks & Libraries

- Pandas, NumPy, Scikit-learn
- PyTorch, TensorFlow, Matplotlib, Seaborn
- Flask / FastAPI
- Hugging Face Transformers

Web Programming (Supporting Skills)

- HTML5, CSS3, JavaScript
- Responsive UI design (Bootstrap/Tailwind)
- Node.js & Express.js (backend development)
- Docker

Database Management Systems

- DBMS/RDBMS fundamentals (normalization, indexing, transactions)
- MySQL (queries, joins, aggregations)
- PostgreSQL (relational database design)
- MongoDB (Document-based data modeling)

Programming Tools

- Git & GitHub
- VS Code / Jupyter Notebook
- Docker
- MS Excel

Programming Language

- Python (Primary)
- SQL, JavaScript, HTML, CSS (Supporting)

Soft Skills

- Problem-solving and Analytical Thinking
- Attention to Detail
- Flexibility and Active Learning
- Communication & Presentation
- Adaptability & Team Collaboration

Experience

Technetics IT Services – AI/ML Intern *Jan 2026 – Present | Surat, Gujarat, India*

- Built a **Retrieval-Augmented Generation (RAG) based document question-answering system**, implementing text preprocessing, chunking, and semantic search with vector embeddings to generate context-aware responses and improve retrieval relevance.
- Integrated trained ML models into a **FastAPI backend** to expose prediction APIs, and utilized **Docker for containerized deployment** to ensure consistent and scalable development environments.
- Developed a **Variational Autoencoder (VAE)** for feature learning and basic anomaly detection, enhancing data representation quality and evaluating performance using accuracy, reconstruction loss, and precision@k.

Logicrest – Research Intern (Hardware Team Lead) *Sep 2025 – Dec 2025 | Surat, Gujarat, India (Remote)*

- Led the **hardware intern team** in researching and selecting industrial-grade sensors for a **Sewage Treatment Plant (STP) monitoring system**, ensuring suitability for real-time, industrial environments.
- Conducted technical analysis of **biological, chemical, and filtration-based water treatment processes**, and evaluated key water quality sensors (pH, turbidity, TDS, dissolved oxygen) for accurate monitoring.
- Studied **microcontroller and PLC-based automation architectures** and assisted in designing a hardware framework to integrate sensors with monitoring and data acquisition systems.

Projects

AI-Powered Resume Analyzer & Career Intelligence Platform - Demo

- Built an **end-to-end AI-powered resume analysis system** delivering Role Matching, ATS Scoring, Skill Gap Detection, and Career Path Recommendations using NLP and semantic similarity techniques.
- Designed a **hybrid role-matching and explainable ATS engine**, combining semantic embeddings, skill overlap, experience alignment, keyword relevance, and category-wise scoring for job-fit evaluation.
- Developed **skill normalization and fuzzy matching pipelines** with an interactive multi-tab dashboard (radar charts, ranked insights), enabling structured report generation and actionable career intelligence.

AI-Powered Logistics Intelligence & Command System - Demo

- Designed a **human-in-the-loop AI-powered logistics decision support system** for pre-dispatch shipment risk assessment using multi-factor analysis, structured validation pipelines, and real-time weather risk integration.
- Developed a **hybrid Rule-Based + Decision Tree Classifier model** to evaluate route complexity, vehicle feasibility, urgency, and weather impact, optimizing risk scoring logic for clear decision-making.
- Built **role-based Streamlit dashboards (Seller, Manager, Supervisor)** with audit logging, override tracking, explainable AI outputs, and a complete decision audit trail to ensure governance, traceability, and operational transparency.

Education

Bachelor of Engineering

C.K. Pithawalla College of Engineering & Technology – Surat, Gujarat, India

Higher Secondary Certificate (HSC) in Science Stream

Ashadeep IIT – Surat, Gujarat [2023]

Secondary School Certificate (SSC)

K. & M. P. Sarvajanik Vidhyalay – Surat, Gujarat [2021]

Certifications & Achievements

- IBM - Python for Data Science, AI & Development - Credential
- Oracle Cloud Infrastructure 2025 Certified Data Science Professional - Credential
- Google - Crash Course on Python - Credential
- Google - Regression Analysis: Simplify Complex Data Relationships - Credential

Interests & Activities

- Learning about Generative AI and emerging ML trends through articles, tutorials, and research papers.
- Participating in Technical discussions and sharing knowledge with peers.
- Studying automation and control systems used in smart infrastructure and industrial operations.
- Experimenting with Automation tools and improving workflow efficiency